Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Foundations of Data Analytics				
Course Code:	ANA1000	Credit Value:	3	Credit Hours:	42
Programs:	BAPG Business Analytics CAGC Crime Analytics HAGC Health Analytics				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐ Degree Breadth Course: ☐ Eligible for PLAR: ☑

COURSE DESCRIPTION

In this course, students will explore the growth of data analytics and its role in strategic decision-making processes. Students will discover data science and some of the key tools used in analytics and the purpose of each tool in the collection, storage, management, and communication of data to various stakeholders. Students will also investigate the ethical and legal requirements of working with and communicating data.

Date: May 25, 2021

Prepared By: Sidney Shapiro

Approved by: Brian Lobban

BRIAN LOBBAN
Dean, School Of Business And Information Technology, Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL

This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:

Program(s)	Vocational Learning Outcomes
Business Analytics	**College Standards** 1. Collect, manipulate and mine data sets to meet organizational need. 2. Design and apply data models that meet the needs of a specific operational/ business process. 3. Design and present data visualizations to communicate information to business stakeholders. 4. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.
Crime Analytics	**College Standards** 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Design and present data visualizations to communicate information to stakeholders.
Health Analytics	**College Standards** 1. Design health indicators and apply data models that meet organizational, program and service needs. 2. Apply health analytics and business intelligence tools to support evidence-informed practice. 3. Assess organizational requirements for health information needs and evaluate the impact on the organization, programs and services, and on health services delivery. 4. Synthesize relevant local, regional, provincial, national and global health and health information management issues, trends, technologies and standards to support health information systems and processes. 5. Ensure the completeness, accuracy, consistency, timeliness and integrity of health information throughout the management of its lifecycle.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- The demand for business intelligence
- Data and statistics
- Data warehousing
- Data mining processes and technologies
- Analyzing text, web, and social media data
- Optimization and decision-making

Topics/Concepts Covered in This Course

- Big data
- Ethical and legal considerations of data analytics

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	**Objectives**
1. Define and describe the basic concepts of data analytics	1.1 Define data analytics and business intelligence, and describe some of the tools they use 1.2 Discuss the history of data analytics 1.3 Compare and contrast the three types of analytics 1.4 Explain some of the key tools used in descriptive, predictive, and prescriptive analytics 1.5 Give examples of how analytics can be used to enhance business performance 1.6 Compare and contrast traditional data analysis with big data analysis 1.7 Identify the different parties involved in data analytics
2. Illustrate the basic methods of descriptive analytics	2.1 Describe the various types of data 2.2 Identify characteristics of data that is suitable for analytics describe data preprocessing methods 2.3 Calculate basic statistics including the mean, median, mode, range, standard deviation 2.4 Interpret a boxplot and histogram 2.5 Identify the skew and kurtosis of a distribution, and their accompanying characteristics 2.6 Discuss the process, use, and assumptions of regression modelling 2.7 Describe the uses of and different types of business reports 2.8 Explain various data visualization techniques, and identify which are appropriate for a given setting
3. Explain the role of data warehouses in analytics	3.1 Define the term "data warehouse" and describe their main characteristics, components, and structure 3.2 Give examples of how data warehouses can support business performance 3.3 Compare and contrast different data warehouse architectures 3.4 Explain the ETL process and characteristics of ETL tools 3.5 Illustrate the different structures for storing data in a warehouse

Learning Outcome	Objectives
	3.6 Compare and contrast OLTP with OLAP 3.7 Identify the OLAP operations required to extract a certain subset of data from a warehouse 3.8 Discuss the issues of scalability and security in data warehousing 3.9 Discuss the issues of scalability and security in data warehousing 3.10 Explain the business performance management process 3.11 Compare and contrast balanced scorecard with sig sigma 3.12 Discuss the relevance of KPIs to business performance
4. Describe standard data mining tools/ techniques and explain their role business analytics	4.1 Define the concept of data mining and list its main characteristics/uses 4.2 Describe the main data mining techniques and their associated algorithms 4.3 Give examples of the uses of data mining in various industries 4.4 Compare and contrast the CRISP-DM, SEMMA, and KDD data mining processes 4.5 Given a confusion matrix, calculate the accuracy, TPR, TNR, Precision, and Recall 4.6 Interpret an ROC curve 4.7 Calculate and interpret the support, confidence, and lift values in association rule mining 4.8 List various commercial and open-source data mining technologies 4.9 Identify common data mining errors and how to avoid them
5. Explain the techniques for text, web, and social media analytics, as well as their role in business intelligence	5.1 Define the concepts of text analytics and text mining, and describe their main characteristics and application areas 5.2 Compare and contrast data mining with text mining 5.3 Define the main terms associated with text mining 5.4 Explain natural language processing, its main tasks/ tools, and its role in text mining 5.5 Explain the three-step text mining process 5.6 Define sentiment analysis, and identify the four-step sentiment analysis process 5.7 Identify some challenges of web mining and the three main categories of web mining 5.8 Compare and contrast the tools, challenges, and applications of web content, structure, and usage mining

Learning Outcome	Objectives
	5.9 Describe the structure of a search engine and explain the technique of search engine optimization 5.10 Identify the key web analytics metrics 5.11 Describe the key metrics for social media analytics
6. Identify how optimization and simulation can be utilized in a business setting	6.1 Define prescriptive analytics and some of the main issues regarding model-based decision making 6.2 List the different prescriptive modelling techniques, their processes, and objectives 6.3 Compare and contrast model-based decision making under certainty and uncertainty 6.4 Describe the characteristics of spreadsheet modeling 6.5 Describe linear programming, its uses, characteristics, and steps 6.6 Formulate a problem as a linear program 6.7 Interpret the solution of a linear program 6.8 Give examples of common optimization problems 6.9 Describe methods of decision-making with multiple goals and some of the related challenges 6.10 Explain the concepts of sensitivity analysis, what-if analysis, and goal-seeking and what they are used for 6.11 Given a situation, create a decision table and calculate the optimal decision 6.12 Explain the characteristics of simulation, its process, when it should be used, its advantages and disadvantages, types of simulation, and common simulation software
7. Explain the difficulties of big data analytics, as well as the technological innovations that have been designed to overcome them	7.1 Identify the main characteristics, requirements, and enablers of big data analytics 7.2 List and explain the main challenges of big data analytics 7.3 Define data governance and describe the importance of data governance in a big data context 7.4 Identify the main big data technologies and their functions 7.5 Explain how MapReduce distributes processing, as well as some of its pros and cons 7.6 Explain how Hadoop works and identify its main technical components 7.7 Compare contrast data warehousing with Hadoop 7.8 Explain the purpose and architecture of the big data platforms IBM InfoSphere and Teradata Aster 7.9 Explain why stream analytics is essential in some industries

Learning Outcome	Objectives
	7.10 Give examples of how stream analytics can be used in various organizations
8. Review the current trends in data analytics, and summarize the ethical and legal considerations of working with big data	8.1 Define the Internet of Things and give reasons for its growth 8.2 Explain the characteristics and function of RFID sensors, and their role in the Internet of Things 8.3 Compare and contrast fog computing with cloud computing 8.4 Define and give examples of cloud computing 8.5 Compare and contrast the different service-oriented cloud computing architectures 8.6 Describe geospatial and location-based analytics, and the relevance of geographic information systems to this field 8.7 Consider the different legal, privacy, and ethical issues surrounding data analytics, and the possible consequences of these issues for the parties involved 8.8 Explain the impact of data analytics on the operation and structure of organizations

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.

Interpersonal

- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Self-Directed Learning
- Presentations
- In-Class Exercises
- Case Studies
- eLearning Components

Resources Required

Books

Sharda, Delen, & Turban, *Business Intelligence, Analytics, and Data Science: A Managerial Perspective*, 4th, Pearson
ISBN: 0134633288

Evaluation Plan

Grading Scheme

A	80%	-	100%
B	70%	-	79%
C	60%	-	69%
D	50%	-	59%
F	0%	-	49%

Evaluation Method	Value (%)
Applied Activities (G)	**10%**
For selected modules, an article will be given detailing a real-world application or issue concerning data analytics. Students will read the article, and then answer the given questions.	
Assignments	**60%**
Students will choose from a list provided by the instructor and write a report outlining how these companies can implement data analytical tools to improve their businesses practices. Students will write an essay on the ethical and legal considerations of working with big data. Relevant news articles or discussion questions will be posted on the class forum throughout the course. Each student must write a thoughtful response to each article/question as assigned by the instructor.	
Quizzes	**30%**
There will be quizzes during the term as determined by the instructor. Each quiz is equally weighted.	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan
Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies
Students must adhere to all academic policies. These are available on myCambrian as well as the College website at https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/.

Accessibility Services
Cambrian is committed to creating an inclusive learning environment where we support our students" success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance
Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/ participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture
Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright
Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: http://cambriancollege.libguides.com/copyrightinfo or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services
Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance
The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)
Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian
Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations
Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit
Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity
Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/